SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 3– Hackathon

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 – Hackathon**	**Assessment:**	**Assessment Tool 3– Hackathon**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	Jothika M	**Reg. No.:**	192324235
Section / Batch:	2023-B.Tech (AIDS)	**Date:**	29/08/2026

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jothika M

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

1. Disease Symptom Checker. Predict likely disease category from a patient's symptom vector (fever, cough, fatigue, etc.) using kNN with different values of k and distance metrics (Euclidean vs Manhattan).

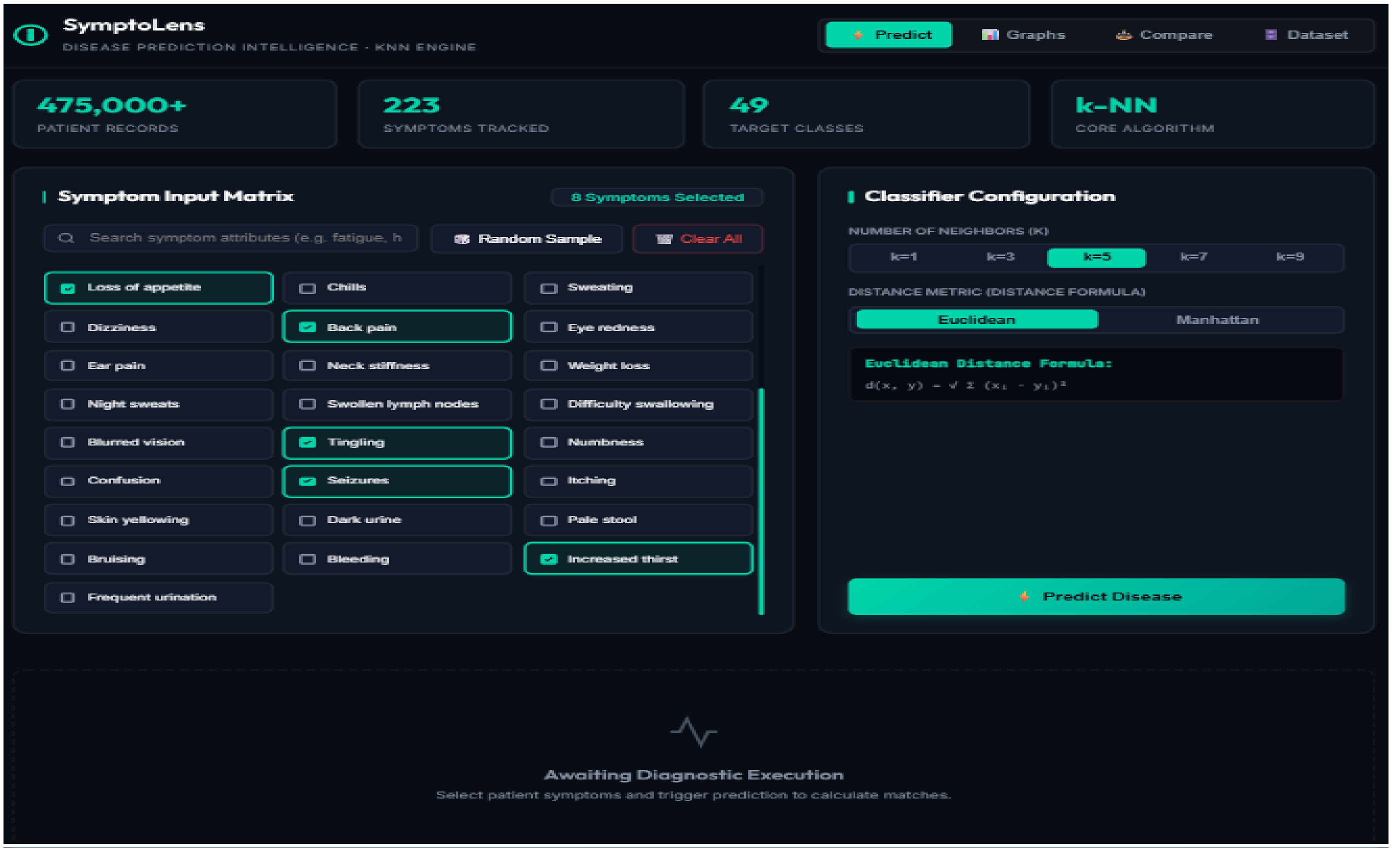
Problem Statement:

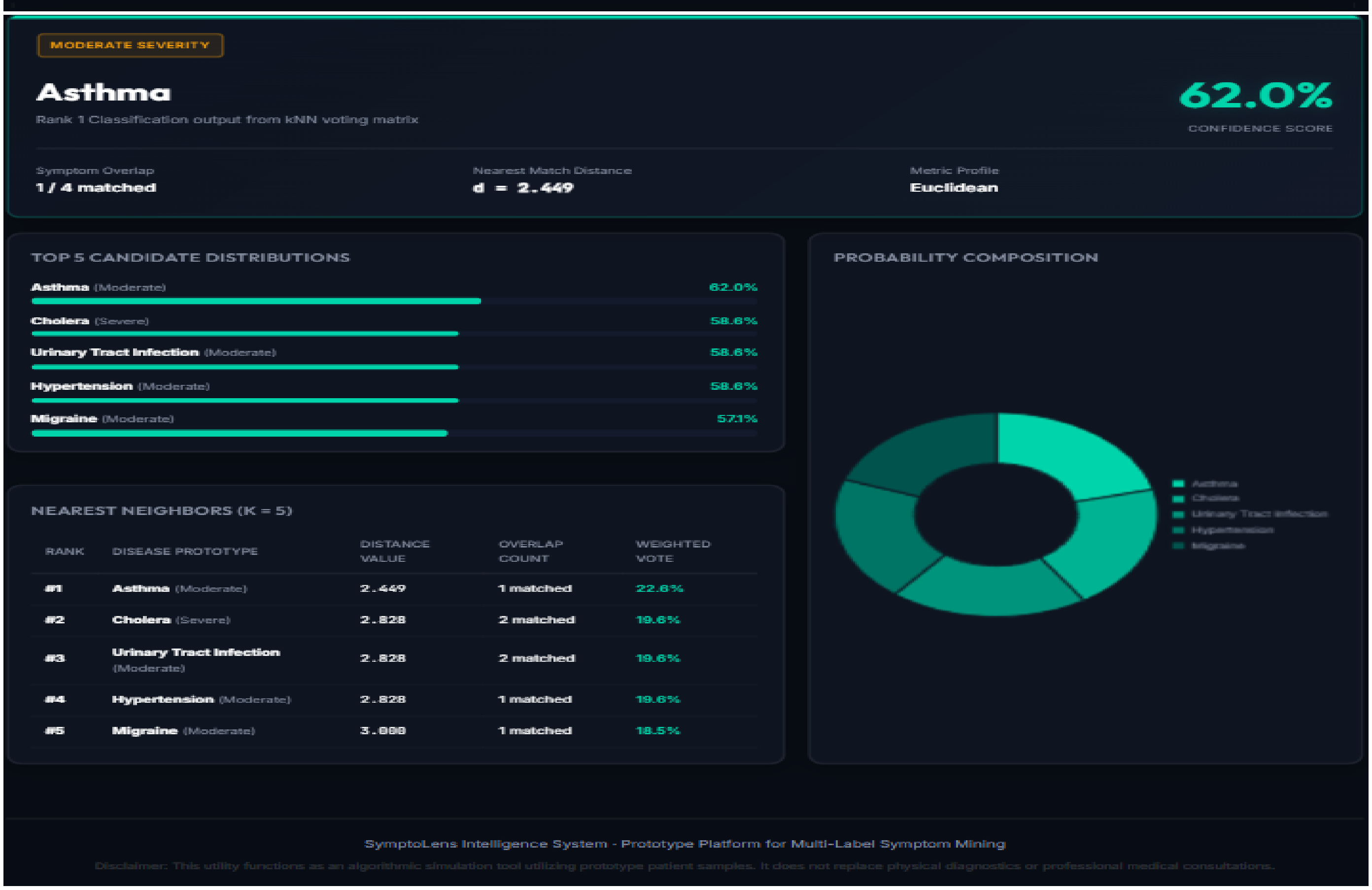
Develop a Disease Symptom Checker that predicts the likely disease category of a patient based on their symptoms such as fever, cough, fatigue, headache, and body pain. The system should use the k-Nearest Neighbors (kNN) classification algorithm to classify patients into different disease categories. Evaluate the performance of kNN using different values of k and compare Euclidean and Manhattan distance metrics to determine which combination provides better prediction accuracy.

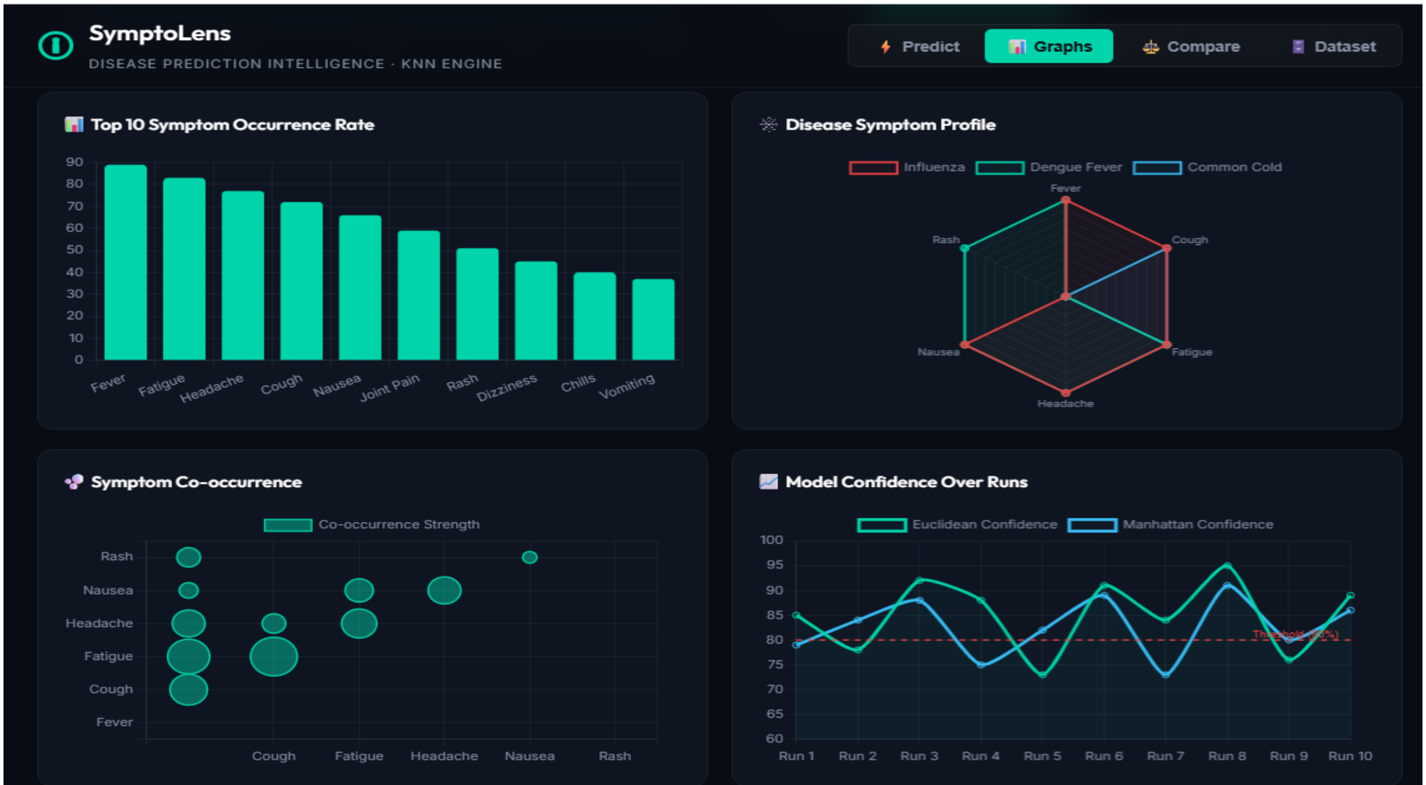
Code:

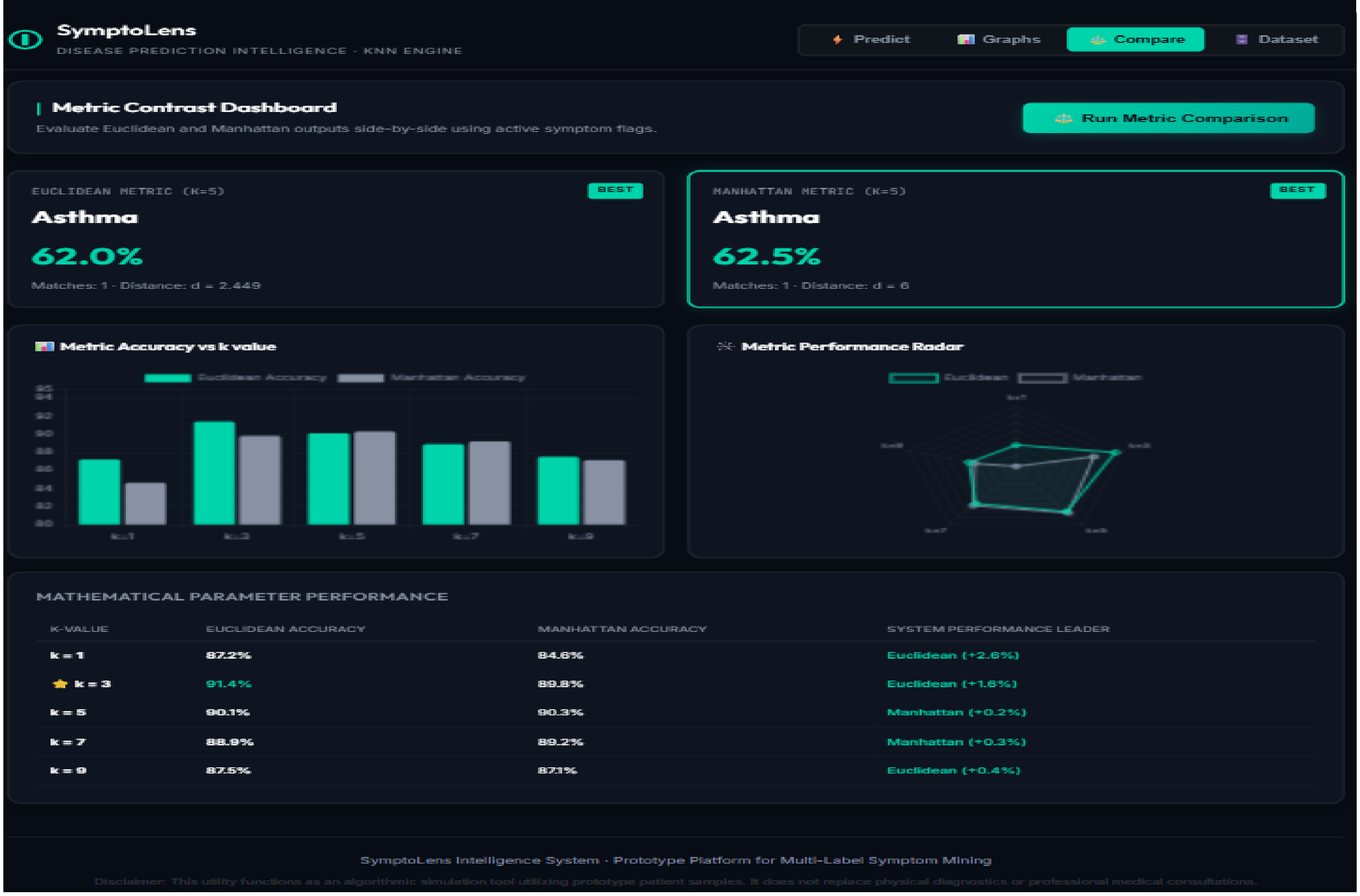
```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SymptoLens — Disease Prediction Intelligence</title>

  <!-- Google Fonts: Inter & Outfit, JetBrains Mono for monospace expressions -->
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&family=Outfit:wght@400;500;600;700&family=JetBrains+Mono:wght@

  <!-- Chart.js 4.4.1 UMD Bundle -->
  <script src="https://cdnjs.cloudflare.com/ajax/libs/Chart.js/4.4.1/chart.umd.js"></script>

  <style>
    /* CSS Variables & Theme Configuration */
    :root {
      --bg-color: #080c14;
      --card-bg: #0e1420;
      --card-hover: #141d2f;
      --accent-color: #00d4aa;
      --accent-glow: rgba(0, 212, 170, 0.2);
      --accent-muted: #008f72;
      --input-bg: #121a2b;
      --text-main: #ffffff;
      --text-muted: #8892a4;
      --text-dark: #080c14;
      --border-color: rgba(255, 255, 255, 0.06);
      --border-hover: rgba(255, 255, 255, 0.12);

```

Output:


SymptoLens
DISEASE PREDICTION INTELLIGENCE · KNN ENGINE
Predict
Graphs
Compare
Dataset
475,000+
PATIENT RECORDS
223
SYMPTOMS TRACKED
49
TARGET CLASSES
k-NN
CORE ALGORITHM
Symptom Input Matrix
8 Symptoms Selected
Search symptom attributes (e.g. fatigue, h
Random Sample
Clear All
Loss of appetite
Chills
Sweating
Dizziness
Back pain
Eye redness
Ear pain
Neck stiffness
Weight loss
Night sweats
Swollen lymph nodes
Difficulty swallowing
Blurred vision
Tingling
Numbness
Confusion
Seizures
Itching
Skin yellowing
Dark urine
Pale stool
Bruising
Bleeding
Increased thirst
Frequent urination
Classifier Configuration
NUMBER OF NEIGHBORS (K)
k=1
k=3
k=5
k=7
k=9
DISTANCE METRIC (DISTANCE FORMULA)
Euclidean
Manhattan
Euclidean Distance Formula:
Predict Disease
Awaiting Diagnostic Execution
Select patient symptoms and trigger prediction to calculate matches.


MODERATE SEVERITY
Asthma
Rank 1 Classification output from kNN voting matrix
62.0%
CONFIDENCE SCORE
Symptom Overlap
1 / 4 matched
Nearest Match Distance
d = 2.449
Metric Profile
Euclidean
TOP 5 CANDIDATE DISTRIBUTIONS
Asthma (Moderate) 62.0%
Cholera (Severe) 58.6%
Urinary Tract Infection (Moderate) 58.6%
Hypertension (Moderate) 58.6%
Migraine (Moderate) 57.1%
PROBABILITY COMPOSITION
Asthma
Cholera
Urinary Tract Infection
Hypertension
Migraine
NEAREST NEIGHBORS (K = 5)
RANK
DISEASE PROTOTYPE
DISTANCE VALUE
OVERLAP COUNT
WEIGHTED VOTE
#1 Asthma (Moderate) 2.449 1 matched 22.6%
#2 Cholera (Severe) 2.828 2 matched 19.6%
#3 Urinary Tract Infection (Moderate) 2.828 2 matched 19.6%
#4 Hypertension (Moderate) 2.828 1 matched 19.6%
#5 Migraine (Moderate) 3.000 1 matched 18.5%
SymptoLens Intelligence System - Prototype Platform for Multi-Label Symptom Mining
Disclaimer: This utility functions as an algorithmic simulation tool utilizing prototype patient samples. It does not replace physical diagnostics or professional medical consultations.





K-VALUE	EUCLIDEAN ACCURACY	MANHATTAN ACCURACY	SYSTEM PERFORMANCE LEADER
k = 1	87.2%	84.6%	Euclidean (+2.6%)
⭐ k = 3	91.4%	89.8%	Euclidean (+1.6%)
k = 5	90.1%	90.3%	Manhattan (+0.2%)
k = 7	88.9%	89.2%	Manhattan (+0.3%)
k = 9	87.5%	87.1%	Euclidean (+0.4%)

KAGGLE SOURCE

Disease Symptom Prediction Dataset

Published by author **Muhereza Alouzious.** Features categorical representations of clinical cases designed for multi-label classification engines.

Open Dataset on Kaggle →

SOURCE DATABASE	PATIENT COHORT	TOTAL ATTRIBUTES	TOTAL CLASSES
Kaggle (Muhereza A.)	**475,000+ Cases**	**223 Features**	**49 Diseases**
EXPORT FILE FORMAT	MISSING VALUES	LABEL CLASSIFICATION	FEATURE VECTOR FORMAT
CSV Matrix	**0% (Null Free)**	**Categorical String**	**Binary Flags [0/1]**

Embedded Dataset Overview (30 Prototype Vectors)

Download CSV   Download JSON

DISEASE NAME	FEVER	COUGH	FATIGUE	HEADACHE	NAUSEA	VOMITING	DIARRHEA	CHEST PAIN	SHORTNESS OF BREATH	SORE THROAT
Common Cold	0	1	1	0	0	0	0	0	0	1
Influenza	1	1	1	1	1	0	0	0	0	1
COVID-19	1	1	1	1	0	0	0	1	1	1
Pneumonia	1	1	1	1	0	0	0	1	1	0
Tuberculosis	1	1	1	1	0	0	0	1	1	0
Malaria	1	0	1	1	1	1	0	0	0	0
Dengue Fever	1	0	1	1	1	1	0	0	0	0
Typhoid	1	0	1	1	1	1	1	0	0	0
Hepatitis A	1	0	1	1	1	1	0	0	0	0
Hepatitis B	1	0	1	1	1	1	0	0	0	0

FULL KAGGLE DATASET TARGET DISEASES (49 TOTAL)

Common Cold · Influenza · COVID-19 · Pneumonia · Tuberculosis · Malaria · Dengue Fever · Typhoid · Hepatitis A · Hepatitis B · Hepatitis C · Hepatitis D · Hepatitis E · Diabetes · Hypertension · Asthma · Anemia · Kidney Disease · Meningitis · Appendicitis · Gastroenteritis · Migraine · Cholera · Measles · Chickenpox · Urinary Tract Infection · Tonsillitis · Sinusitis · Bronchitis · Food Poisoning · Lupus · HIV/AIDS · Arthritis · Osteoarthritis · Psoriasis · Gout · Hypothyroidism · Hyperthyroidism · Hypoglycemia · Stroke · Heart Attack · Acne · Eczema · Allergy · GERD · Peptic Ulcer Disease · Hemorrhoids · Varicose Veins · Paralysis

Geotag :


GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.02644° Long 80.016927°
Saturday, 29/08/2026 08:53 AM GMT +05:30
Google